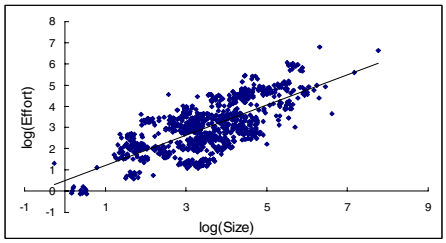
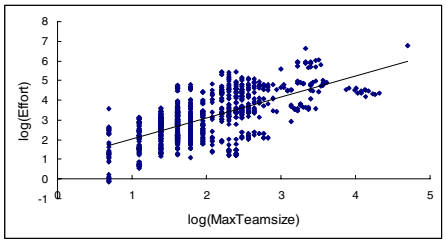
After that, the potential relationships between Effort and the factors (Project Size, Team Size and Duration) after log transformation are explored. The three graphs below (see Fig. 2) indicate that linear model can be used to approximate their relationships with effort. A multiple linear regression can be applied to develop our model. The linear model is supposed to be in form of:

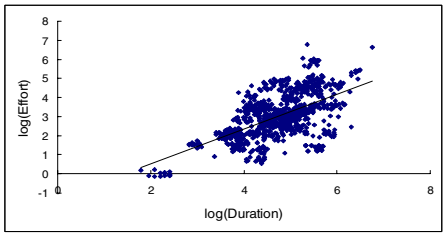
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_k X_k + \mu \quad (2)$$



(a). log(Effort) against log(Size)



(b). log(Effort) against log(TeamSize)



(c). log(Effort) against log(Duration)

Fig. 2. Scatter plots of effort against factors

Furthermore, the correlation analysis is employed to check whether the problem of multi-colliearity (strong correlations between independent variables) exists in the data. As recommended by Maxwell [12], Spearman's correlation analysis is done to check the numerical variables' independence; ANOVA (analysis of variance) is run to check the independence between the categorical variables and chi-square test for the relationship between the categorical and numerical variables. The result confirms that multi-colliearity within this data is not a problem.

In the modeling procedure, three numerical variables, Project Size, Team Size and Duration, passed the check and can be added into one model; but there exist some correlations between any two of the categorical variables, i.e. DevType, BusiArea and Language.

In addition, to explore the problem of missing values, the metrics with missing data are Duration (22), Development Type (13), and Team Size (1). According to the rule of thumb, a minimum sample size of 50+8k for multiple regression analysis is suggested [13]. The valid sample size here is acceptable.

Once the above issues are solved, the regression model can be developed by following the two steps recommended in [12]. At the same time, we also use the statistical tool (Stata [19]) to assist our analysis.